AI A2 Technical Report - Musanze Harvest Intelligence Service

1. Identity and decision problem

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Personal random seed LLLL: 7483

Stakeholder: a potato cooperative in Musanze whose field officer receives farm records in inconsistent spreadsheets and needs an evidence-based decision-support tool.

Prediction questions:

- Q1 (regression): What harvest quantity (yield_tons) should this farm configuration produce?
- Q2 (classification): Is this farm at high production risk (yield per hectare below the dataset median)?
- Q3 (clustering): Which group of farms with similar conditions does this farm belong to?

Ethical boundary: this is a classroom decision-support prototype on synthetic data. It must not replace an agronomist and must not determine access to finance, insurance or public services. Every prediction requires human review by a qualified field officer.

2. Data and reproducibility

The dataset is fully synthetic, generated by data/generate_data.py with numpy.random.default_rng(7483). Features: rainfall_mm U(600,1400), avg_temp_c U(12,26), soil_moisture_pct U(20,80), fertilizer_kg U(0,250), field_size_ha U(0.2,4.0), season in {A,B}, cooperative_support in {0,1}. Target: yield_tons from the course formula (linear effects of rainfall, moisture, fertilizer, field size, temperature deviation from 18C, support, season effect +0.50/-0.20, Gaussian noise sigma=0.80, clipped at 0.50). high_risk = 1 when yield per hectare is below its median.

Row count: base rows = 240 + (7483 mod 41) = 261. Deliberate defects: m = round(0.05 x 261) = 13 missing values injected into soil_moisture_pct and 13 into fertilizer_kg (chosen by the same generator), plus 3 exact duplicate rows appended; raw CSV therefore has 264 rows. One duplicated row also carried a missing fertilizer value, so validation reports 14 missing fertilizer entries in the raw file. Cleaning removed the 3 duplicates, leaving 261 unique rows. Numeric values are stored with float_format='%.4f'.

SHA-256 dataset fingerprint: b9a0067604ae61a7add8d7e174a95e588645899008b97feac17f468bb1b8aff2

Reproducibility: re-running run_all.py --student-id ug2527483@ines.ac.rw --seed 7483 regenerates the identical CSV fingerprint and identical metrics (verified twice in this build).

Train/test split

Fixed 80/20 train/test split with random_state=7483 (208 train / 53 test rows after cleaning). No imputer, scaler, encoder or feature selector ever sees the test set: all transformers live inside sklearn Pipelines fit only on training data.

3. Pipeline design

Modules under src/: config.py (seed, paths, features), validation.py (seed check, schema, ranges, missing/duplicate/leakage checks, SHA-256 fingerprint), preprocessing.py (cleaning + ColumnTransformer + split), regression.py, classification.py, clustering.py, artifacts.py (JSON/CSV/PNG persistence and plots). Entry point run_all.py orchestrates all seven stages and prints the verification code throughout.

Preprocessing flow

- Numeric features: median imputation + StandardScaler (fit on train only).
- Binary cooperative_support: median imputation only.
- Categorical season: most-frequent imputation + one-hot encoding (handle_unknown=ignore).

Leak

- yield_tons, yield-per-hectare and high_risk are never inputs; classification uses only the seven farm-condition features.
- Clusters are fit on scaled condition features only; targets are excluded from clustering_report.json by design.
- Imputers/scalers/encoders are inside the ColumnTransformer inside each Pipeline, so training and inference share identical transformations.

Mode

- Regression: mean baseline, linear regression, gradient boosting (n_estimators default, random_state=7483).
- Classification: majority baseline, logistic regression (max_iter=1000), random forest (200 trees).
- Clustering: KMeans with n_init=10, random_state=7483, K tested from 2 to 6, K chosen by silhouette score.

4. Regression and classification results

Regression (target yield_tons, 53 held-out test rows)

Model	MAE	RMSE	R2
mean_baseline	2.1694	2.7027	-0.0285
linear_regression	0.876	1.0765	0.8368
gradient_boosting	0.9343	1.2294	0.7872

Selected model: linear_regression. Selected linear_regression because it achieved the lowest test MAE (0.876) among 3 candidates, with RMSE 1.0765 and R2 0.8368. The linear model matches the formula-based generator better than gradient boosting, which slightly overfits the small training set.

Residual interpretation (one important error): residual plot artifacts/plots/residuals.png shows errors centred near zero with no strong funnel shape; larger absolute errors occur for large-field farms because field_size_ha has the biggest coefficient (1.30 t/ha), so a modest relative miss on that feature amplifies the tonnage error. Mean baseline R2 is negative (-0.0285), confirming the test set is not trivially predictable by the training mean.

Classification (target high_risk)

Model	Prec	Recall	F1	AUC	CM [[TN,FP],[FN,TP]]
majority_baseline	0.0	0.0	0.0	0.5	[[26,0],[27,0]]
logistic_regression	0.9643	1.0	0.9818	1.0	[[25,1],[0,27]]
random_forest	0.9615	0.9259	0.9434	0.9858	[[25,1],[2,25]]

Selected model: logistic_regression. Selected logistic_regression because it achieved the highest test F1 (0.9818) among 3 candidates, with precision 0.9643, recall 1.0 and ROC-AUC 1.0. The majority baseline scores F1=0 and AUC=0.5 because it never predicts the positive class (26 true negatives, 27 false negatives).

One classification error: logistic regression produced a single false positive (actual low risk predicted high risk, CM [[25,1],[0,27]]). That farm's features sit close to the median-yield-per-hectare decision boundary, so a small feature-side shift flips the label; recall 1.0 means no high-risk farm was missed, which is the safer error direction for a field officer screening for risk.

5. Clustering

K	Silhouette
2	0.171
3	0.1447
4	0.1471
5	0.1469
6	0.1545

Selected K: 2. K=2 maximised the silhouette score (0.171) across K=2..6 on scaled farm-condition features. Silhouette scores are modest (0.14-0.17) because the synthetic features are drawn from broad independent uniforms, so clusters are weak by construction; the relative maximum at K=2 is still the defensible choice.

Cluster profiles (targets excluded)

Cluster 0 (n=132): median rainfall 993.2781 mm, soil moisture 49.6449%, fertilizer 124.0647 kg, field size 1.9199 ha, cooperative_support = 1. A cluster of 132 farms. Rainfall is generally low to moderate. Soil moisture tends to be limited. Fertilizer use is relatively light. Fields are on the smaller side. Most farms receive cooperative support. Operational meaning: supported, moderate-input farms; a natural group for cooperative extension visits.

Cluster 1 (n=129): median rainfall 1027.0239 mm, soil moisture 51.7366%, fertilizer 119.3594 kg, field size 1.8015 ha, cooperative_support = 0. A cluster of 129 farms. Rainfall is generally low to moderate. Soil moisture tends to be abundant. Fertilizer use is relatively light. Fields are on the smaller side. Most farms do not receive cooperative support. Operational meaning: unsupported farms with slightly higher moisture; a candidate group for support-outreach programmes.

One cluster interpretation: cluster 0 is defined mainly by cooperative_support = 1 at its median; the condition features are close to overall medians, so the algorithm split the population on the only strongly bimodal input (the 0/1 support flag). The field officer should read cluster membership as a support-status segmentation, not an agronomic zone.

6. API and testing

Endpoint contract

- GET /health -> {status: ok, verification_code: AI-A2-7483}
- GET /model-info -> student, seed, selected models, K, features, fingerprint, intended use
- POST /predict -> typed FarmRecord body; returns verification_code, yield_tons, high_risk, high_risk_probability, cluster

Valid example (samples/valid.json): rainfall 1000 mm, temp 18 C, moisture 45%, fertilizer 120 kg, field 1.5 ha, season A, support 1 -> yield_tons 11.2163, high_risk 0, probability 0.12, cluster 0.

Invalid example: out-of-range rainfall (50 mm) or missing fields returns HTTP 422 with a structured Pydantic error listing the failing field, expected bounds and input value. Wrong types (e.g. fertilizer_kg = 'lots') and unknown season values are rejected the same way.

Automated tests (pytest -q: 9 passed)

- test_seed_validation_rejects_wrong_seed / test_seed_validation_accepts_personal_seed - seed rule
- test_dataset_reproducible - identical SHA-256 fingerprint from two independent generations
- test_valid_inference - full inference path returns all required fields and verification code
- test_missing_field_rejected - missing rainfall_mm detected
- test_wrong_type_rejected - non-numeric fertilizer_kg detected
- test_range_error_rejected - rainfall_mm=50 rejected by range rule
- test_invalid_season_rejected - season 'C' rejected
- test_api_health_and_predict - FastAPI /health, valid /predict (200) and invalid /predict (4xx)

Failure behaviour: predict.py exits with code 1 and lists every validation error; the API returns 422 with machine-readable details; if models are missing the API returns 503 with instructions to run run_all.py.

7. Model card and reflection

Intended use

Classroom decision-support prototype illustrating yield estimation, risk screening and farm segmentation for a Musanze potato cooperative demo on synthetic data.

Limitations

- Targets follow a linear formula with Gaussian noise; real agronomy adds pests, soil type, timing and market effects.
- 261 unique rows and 53 test rows give wide metric uncertainty; ROC-AUC 1.0 reflects the easy synthetic boundary, not real-world perfection.
- Weak silhouette scores show the uniform feature draws do not form natural agronomic zones.

Risks

- Must not replace an agronomist or automated field diagnostics.
- Must not determine access to credit, insurance, subsidies or any public service.
- Must not be presented as measured Musanze field data; the dataset is 100% synthetic.

Huma

Every prediction must be reviewed by a qualified field officer or agronomist before any operational decision. The API and CLI both print the verification code AI-A2-7483 so outputs can be traced to this submission.

What I would improve with real data

- Replace synthetic targets with measured yields and add pest, soil and management-history features.
- Use cross-validation instead of a single split, and calibrate class probabilities.
- Revisit clustering with domain features; silhouette-based K would likely move beyond 2.
- Add drift monitoring and a feedback loop from field-officer corrections.